

## GO\_Biological\_Process\_2023 Top pathways by non-permutation

| Geneset                                    | stat        | num.genes | pval      | p.adj     | gene.vals                                                                                |
|--------------------------------------------|-------------|-----------|-----------|-----------|------------------------------------------------------------------------------------------|
| 'De Novo' AMP Biosynthetic Process (GO.0   | -0.08255843 | 4         | 5.674e-01 | 9.886e-01 | PFAS:111 PAICS:7003.5 ATIC:7003.5 ADSS1:7003.5 NA NA                                     |
| 'De Novo' Post-Translational Protein Fol   | -0.13105097 | 15        | 7.895e-02 | 9.886e-01 | DNAJB13:142 SDF2L1:644 HSPA8B:799 HSPA14:923 PTGES3:5662.5 HSPA2:5662.5                  |
| 2--Oxoglutarate-Mediated Processes (GO:000 | 0.08438378  | 13        | 2.922e-01 | 9.886e-01 | GOT1:14 OGDHL:660 GGT2:675 IDH1:5662.5 PHYH:5662.5 ADHF1:5662.5                          |
| 3--UTR-mediated mRNA Destabilization (GO   | -0.02413949 | 11        | 7.816e-01 | 9.886e-01 | DND1:1094 ZFP36:2084 TRIM71:7003.5 RCH31:7003.5 ZFP36L1:7003.5 ZFP36L2:7003.5            |
| 3--UTR-mediated mRNA Stabilization (GO.0   | -0.06303912 | 11        | 4.692e-01 | 9.886e-01 | ELAVL4:1331 YBX3:1592 MYD88:2044 ZFP36:2084 TARDBP:7003.5 MAPK14:7003.5                  |
| 3--Phosphoadenosine 5--Phosphosulfate Me   | 0.10660456  | 8         | 2.965e-01 | 9.886e-01 | SULT1A2:323 PAPSS2:5662.5 TPST2:5662.5 SULT2B1:5662.5 SULT1B1:5662.5 SULT1E1:5662.5      |
| 5S Class rRNA Transcription By RNA Polym   | -0.23755463 | 5         | 6.584e-02 | 9.886e-01 | GTF3C5:373 GTF3C3:578 GTF3C4:1652 GTF3C2:7003.5 GTF3C6:7003.5 NA                         |
| 7-Methylguanosine RNA Capping (GO:000945   | 0.26546942  | 2         | 1.935e-01 | 9.886e-01 | RMNT31:311 CMTR2:5662.5 NA NA NA NA                                                      |
| 7-Methylguanosine mRNA Capping (GO:000663  | 0.26546942  | 2         | 1.935e-01 | 9.886e-01 | RMNT31:311 CMTR2:5662.5 NA NA NA NA                                                      |
| ADP Transport (GO:0015866)                 | -0.12074398 | 5         | 3.498e-01 | 9.886e-01 | SLC25A24:813 SLC25A23:2170 SLC25A42:7003.5 SLC25A41:7003.5 SLC25A5:7003.5 NA             |
| AMP Biosynthetic Process (GO:0006167)      | -0.11099817 | 6         | 3.646e-01 | 9.886e-01 | PFAS:111 APRT:1396 PAICS:7003.5 ATIC:7003.5 ADSS1:7003.5 NUDT2:7003.5                    |
| AMP Metabolic Process (GO:0046033)         | 0.04567587  | 9         | 6.352e-01 | 9.886e-01 | AK9:324 ADSS1:5662.5 NUDT2:5662.5 NT5C1A:5662.5 NT5E:5662.5 AMPD1:5662.5                 |
| ATF6-mediated Unfolded Protein Response    | 0.05376563  | 5         | 6.772e-01 | 9.886e-01 | MBTPS1:5662.5 XBP1:5662.5 ATF6B:5662.5 DDIT3:5662.5 CREBZF:5662.5 NA                     |
| ATP Biosynthetic Process (GO:0006750)      | 0.07835106  | 13        | 3.281e-01 | 9.886e-01 | ATP5F1C:339 ATP5PD:423 NUDT2:5662.5 TGFBI:5662.5 VPS9D1:5662.5 ATP5PF:5662.5             |
| ATP Metabolic Process (GO:0046034)         | -0.02145636 | 25        | 7.105e-01 | 9.886e-01 | PRKAG2:478 NADK:762 PARP1:1028 FIGNL1:1715 MYH8:1858 CTNS:2217                           |
| ATP Synthesis Coupled Electron Transport   | 0.0942873   | 3         | 7.737e-01 | 9.886e-01 | NDUFB6:717 NDUFS2:5662.5 NDUFAV3:10845 NA NA NA                                          |
| ATP Transport (GO:0015867)                 | -0.09707394 | 9         | 3.133e-01 | 9.886e-01 | SLC25A24:813 SLC25A23:2170 SLC25A42:7003.5 SLC25A41:7003.5 SLC25A5:7003.5                |
| AV Node Cell Action Potential (GO:008601)  | -0.12729959 | 6         | 1.439e-01 | 9.886e-01 | CACNB2:335 CALHM1:893 SLC13A2:5662.5 SCN10A:7003.5 SCN5A:7003.5 TRPM4:7003.5             |
| AV Node Cell To Bundle Of His Cell Commu   | 0.05376106  | 4         | 7.096e-01 | 9.886e-01 | SCN5A:5662.5 CXADR:5662.5 GLC3:5662.5 NA                                                 |
| Arp2/3 Complex-Mediated Actin Nucleation   | -0.10453935 | 10        | 6.258e-01 | 9.886e-01 | WASHC1:820 ARPC1A:977 ARPCSL:2053 WHAMM:2179 ARP5C5:7003.5 ARPC3:7003.5                  |
| B Cell Activation (GO:0042113)             | 0.09136123  | 61        | 1.374e-02 | 9.886e-01 | HDAC9:43 CD86:62 FLT3:159 CASP8:207 CD40:419 CD79A:527                                   |
| B Cell Activation Involved In Immune Res   | -0.05942331 | 12        | 4.761e-01 | 9.886e-01 | RNF168:18 MSH2:5662.5 GPR183:5662.5 CD40LG:5662.5 CD180:5662.5 LG4:5662.5                |
| B Cell Differentiation (GO:0030183)        | 0.08932514  | 44        | 4.058e-02 | 9.886e-01 | HDAC9:43 FLT3:159 CD79A:527 PTPRC:780 IL4:881 SLC25A5:5662.5                             |
| B Cell Proliferation (GO:0042100)          | 0.16020800  | 15        | 3.166e-02 | 9.886e-01 | CD40:419 CD179:527 MF2C:761 PTPRC:780 IL4:881 SLC25A5:5662.5                             |
| B Cell Receptor Signaling Pathway (GO:00   | 0.09914482  | 26        | 8.032e-02 | 9.886e-01 | CD79A:527 CTLA4:566 MEFC2:761 PTPRC:780 NFAM1:5662.5 MS4A1:5662.5                        |
| BMP Signaling Pathway (GO:0030509)         | -0.01336705 | 42        | 7.646e-01 | 9.886e-01 | FAM83G:157 INHBE:622 LEFTY1:835 DDX5:960 SMAD5:983 ACVR2A:1596                           |
| C-terminal Protein Amino Acid Modificati   | -0.10195328 | 8         | 3.180e-01 | 9.886e-01 | GPLD1:832 AGBL5:2195 AGBL4:2243 AGTPBP1:7003.5 FOLH1:7003.5 ATG16L1:7003.5               |
| C-terminal Protein Deglutamylation (GO:0   | -0.13526423 | 4         | 3.488e-01 | 9.886e-01 | AGBL5:2195 AGBL4:2243 AGTPBP1:7003.5 FOLH1:7003.5 NA NA                                  |
| C-terminal Protein Lipidation (GO:000655   | 0.09321557  | 3         | 5.120e-01 | 9.886e-01 | GPLD1:832 ATG16L1:7003.5 ATG7:7003.5 NA NA                                               |
| C4-dicarboxylate Transport (GO:0015740)    | 0.03951597  | 10        | 6.653e-01 | 9.886e-01 | SLC25A12:782 SLC25A13:5662.5 SLC13A2:5662.5 SLC25A18:5662.5 SLC13A5:5662.5 SLC1A1:5662.5 |
| CD4-positive, Alpha-Beta T Cell Activati   | -0.04757597 | 5         | 7.126e-01 | 9.886e-01 | NGK7:605 STOML2:7603.5 TNFSF4:7003.5 RSDA2:7003.5 HMG81:7003.5 NA                        |
| CD40 Signaling Pathway (GO:0023035)        | 0.11701448  | 7         | 2.837e-01 | 9.886e-01 | CD86:62 CD40LG:5662.5 ITGB1:5662.5 PHB2:5662.5 TNIP2:5662.5 ITGA5:5662.5                 |
| CENP-A Containing Chromatin Assembly (GO   | 0.04567537  | 5         | 7.236e-01 | 9.886e-01 | MIS18A:728 CENPI:5662.5 OIP5:5662.5 NASP:5662.5 HJURP:11122 NA                           |
| COP1 Coating Of Golgi Vesicle (GO:004820   | 0.05374924  | 2         | 7.923e-01 | 9.886e-01 | GBF1:5662.5 ARFGAP3:5662.5 NA NA NA NA                                                   |
| COP1-coated Vesicle Budding (GO:003964)    | 0.05374924  | 2         | 7.923e-01 | 9.886e-01 | GBF1:5662.5 ARFGAP3:5662.5 NA NA NA NA                                                   |
| COP1 Vesicle Coating (GO:0048208)          | -0.04364555 | 14        | 5.719e-01 | 9.886e-01 | KLHL12:438 TRAPPC11:1323 TRAPPC12:2026 TRAPPC3:7003.5 TRAPPC2L:7003.5 TRAPPC4:7003.5     |
| CRD-mediated mRNA Stabilization (GO:0070   | -0.14997459 | 5         | 2.455e-01 | 9.886e-01 | YBX1:306 IGF2BP1:841 DHX9:7003.5 PABPC1:7003.5 HNRNP:7003.5 NA                           |
| DNA Alkylation (GO:0006305)                | 0.07697969  | 15        | 3.021e-01 | 9.886e-01 | DNMT3B:57 BAZ2A:239 MTRR:5662.5 HELL3:5662.5 TNP2:5662.5 DNMT3A:5662.5                   |
| DNA Catabolic Process (GO:0006308)         | -0.04419696 | 17        | 5.262e-01 | 9.886e-01 | DICER1:391 REXO4:645 DFFB:881 DNASE1L3:900 DNASE1L2:1106 ENDOG:7003.5                    |
| DNA Conformation Change (GO:0071103)       | -0.04860193 | 9         | 6.137e-01 | 9.886e-01 | HMG83:599 TOP3A:1756 TOP2B:7003.5 HMG81:7003.5 TOP2A:7003.5 TOP3B:7003.5                 |

## EnrichmentHsSymbolsFile2 Top pathways by non-permutation

| Geneset                                   | stat         | num.genes | pval      | p.adj     | gene.vals                                                                    |
|-------------------------------------------|--------------|-----------|-----------|-----------|------------------------------------------------------------------------------|
| NIKOLSKY_BREAST_CANCER_20Q12_Q13_AMPLICO  | -0.146877141 | 100       | 4.038e-07 | 2.615e-03 | COL20A1:29 FAM217B:76 MTG2:103 HELZ2:139 CDH26:183 ZMYND8:197                |
| NIKOLSKY_BREAST_CANCER_Q1Q21_AMPLICON     | -0.237384275 | 27        | 1.972e-05 | 6.384e-02 | ADAM15:92 PMVK:1204 RORC:206 MTX1:436 TDRKH:504 ZBTB7B:522                   |
| KEGG_ALLOGRAFT_REJECTION                  | 0.230933275  | 21        | 2.496e-04 | 5.388e-01 | CD86:62 IFNG:93 FASLG:99 IL12B:143 HLA-DQB:232 CD80:296                      |
| PEREZ_TP53_TARGETS                        | -0.036632011 | 812       | 4.740e-04 | 6.139e-01 | GRP143:4 CEMIP:6 APC2:25 KDM2B:26 UNKL:5 SMTNL2:67                           |
| MIKKELSEN_MCV6_HIP_WITH_H3K27ME3          | -0.056807958 | 333       | 3.910e-04 | 6.139e-01 | KCNH2:1 CHRN2:3 CHRNA3:45 HTR1D:78 ZBTB16:99 HTR6:125                        |
| WP_EFFECTS_OF_NCTR_OXIDE                  | -0.395603951 | 6         | 7.912e-04 | 8.540e-01 | NOS3:244 MB:354 AOX1:1500 XDH:1618 NOS1:2017 NOS2:2233                       |
| CHARAFE_BREAST_CANCER_LUMINAL_VS_MESENCH  | -0.053803326 | 319       | 1.018e-03 | 9.419e-01 | EPS8L1:52 SEMA4A:104 FAM83H:117 MLPH:152 KRT19:198 CAPN8:238                 |
| SHARMA_ASTROCYTOMA_WITH_NF1_SYNDROM       | -0.086348307 | 3         | 6.045e-01 | 9.986e-01 | CLEC3B:1708 ADAGR1:7003.5 SLC1A3:7003.5 NA NA NA                             |
| SOGA_COLORECTAL_CANCER_MYC_DN             | 0.038979579  | 56        | 3.134e-01 | 9.986e-01 | AC02:74 SULT1A2:323 SLCA4A:1:614 SULT1A4:684 PLPP1:865 B3GNT5:994            |
| SOGA_COLORECTAL_CANCER_MYC_UP             | 0.028444909  | 63        | 4.353e-01 | 9.986e-01 | GPI:398 UMP5:507 SLC7A6:554 POLR3K:734 DTYMK:985 PAICS:5662.5                |
| CORONEL_RFXY_DIRECT_TARGETS_UP            | 0.014331788  | 23        | 8.120e-01 | 9.986e-01 | XRCC1:905 TSPYL2:1000 ABAT:5662.5 KLF9:5662.5 TOP2B:5662.5 ATRX:5662.5       |
| FOROUTAN_TGFB_EMT_UP                      | -0.008841825 | 138       | 7.205e-01 | 9.986e-01 | SACS:64 MMP9:328 TUF1:407 EML1:464 ARFGAP1:582 PLEK2:594                     |
| FOROUTAN_TGFB_EMT_DN                      | -0.060006315 | 81        | 6.224e-02 | 9.986e-01 | CAVIN2:69 ERMP1:186 KRT19:198 SLC0A1:251 LAMA5:295 JAG2:319                  |
| FOROUTAN_PRODRANK_TGFB_EMT_UP             | -0.017890028 | 132       | 4.788e-01 | 9.986e-01 | SACS:64 MMP9:328 TUF1:407 EML1:464 ARFGAP1:582 PLEK2:594                     |
| FOROUTAN_PRODRANK_TGFB_EMT_DN             | -0.052924699 | 64        | 1.435e-01 | 9.986e-01 | CAVIN2:69 ERMP1:186 KRT19:198 SLC0A1:251 LAMA5:295 JAG2:319                  |
| FOROUTAN_INTEGRATED_TGFB_EMT_UP           | 0.008761891  | 82        | 7.841e-01 | 9.986e-01 | COL1A1:148 VGLL3:427 BMPR2:512 COL6A3:646 ALNLT10:676 SLC26A2:5662.5         |
| FOROUTAN_INTEGRATED_TGFB_EMT_DN           | -0.004839018 | 57        | 2.067e-01 | 9.986e-01 | CAVIN2:69 ERMP1:186 KRT19:198 KRT13:152 ATP8B1:805 BIRC3:885                 |
| GLASS_IGF2BP1_CLIP_TARGETS_KNOCKDOWN_DN   | 0.001230343  | 85        | 6.688e-01 | 9.986e-01 | NCEH1:720 CD2AP:750 SRPK1:882 NABP2:928 SMT2:5662.5 HDLBP:5662.5             |
| HOUNKPE_HOUSEKEEPING_GENES                | 0.001427951  | 694       | 8.993e-01 | 9.986e-01 | RNF168:18 MBD1:58 NARS1:94 CCAR2:126 IPOR:174 PDCD7:178                      |
| IBRAHIM_NRF1_UP                           | 0.009226787  | 275       | 6.002e-01 | 9.986e-01 | DNMT3B:57 MBD1:58 NCKAP1:92 NARS1:94 SCPEP1:149 FUBP1:152                    |
| IBRAHIM_NRF2_UP                           | 0.019770601  | 358       | 2.016e-01 | 9.986e-01 | NCKAP1:92 NARS1:94 TYW3:131 SUPVL3:1224 ZNF189:253 MRPS30:265                |
| IBRAHIM_NRF3_UP                           | 0.053253480  | 2         | 7.942e-01 | 9.986e-01 | HMGX1:5662.5 RPN2:5662.5 NA NA NA NA                                         |
| IBRAHIM_NRF1_DOWN                         | -0.031722755 | 4         | 2.885e-01 | 9.986e-01 | TMEM25:72 KCNH3:95 NPH4A:116 LIG1:243 PIDD1:275 PBXIP1:569                   |
| IBRAHIM_NRF2_DOWN                         | -0.052893290 | 109       | 5.686e-02 | 9.986e-01 | IQGAP3:34 CDK10:102 ENTR1:226 LIG1:243 PPP1R16A:266 PIDD1:275                |
| IBRAHIM_NRF3_DOWN                         | 0.051267514  | 11        | 5.560e-01 | 9.986e-01 | ARHGAP23:700 SMAP1:5662.5 NOG:5662.5 PLAT:5662.5 SEMAC4:5662.5 HECTD3:5662.5 |
| SEAVEY_EPITHELIOID_HEMANGIOENDOTHELIOMA   | 0.040657707  | 67        | 2.503e-01 | 9.986e-01 | C17orf58:205 TNFSF15:272 CDC42EP5:564 NCEH1:720 MYL2:758 PLCD3:5662.5        |
| BENITEZ_GBM_PROTEASOME_INHIBITION_RESPON  | 0.020469099  | 31        | 6.934e-01 | 9.986e-01 | ELP6:978 ASNS:5662.5 MLF2:5662.5 GPATCH2L:5662.5 PNRK1:5662.5 SOX4:5662.5    |
| BLANCO_MELO_SARS_COV_1_INFECTION_MCR5_CE  | 0.053275038  | 7         | 6.255e-01 | 9.986e-01 | KIF14:5662.5 CENPE:5662.5 ASPM:5662.5 ESM1:5662.5 KIF20B:5662.5 BRCA2:5662.5 |
| BLANCO_MELO_SARS_COV_1_INFECTION_MCR5_CE  | 0.173739275  | 4         | 2.289e-01 | 9.986e-01 | HSD17B82:52 CYT11:446 FGFR4:5662.5 RASL12:10386 NA NA                        |
| BLANCO_MELO_MERS_COV_INFECTION_MCR5_CELL  | -0.022826461 | 187       | 2.832e-01 | 9.986e-01 | SUSD4B:168 KCHN:35 RBM15:184 CDKN2AIP:344 ATP2A1:595 FHDC1:650               |
| BLANCO_MELO_MERS_COV_INFECTION_MCR5_CELL  | 0.136302480  | 20        | 3.488e-02 | 9.986e-01 | HSD17B2:52 REX1BD:89 HSD11B1:428 CYT11:446 DNAJC12:581 PNMA2:681             |
| BLANCO_MELO_INFLUENZA_A_INFECTION_A594_C  | 0.065603010  | 26        | 3.205e-01 | 9.986e-01 | CXCL8:629 PLAUR:687 EREG:736 MRPL42:819 CCN2:5662.5 FGFBP1:5662.5            |
| BLANCO_MELO_INFLUENZA_A_INFECTION_A594_C  | -0.043790195 | 78        | 1.817e-01 | 9.986e-01 | APC2:25 OPLAH:364 OVCA2:599 TONSIL:694 SPATA2L:777 TBC1D16:794               |
| BLANCO_MELO_HUMAN_PARAINFLUENZA_VIRUS_3   | 0.0063666834 | 132       | 8.010e-01 | 9.986e-01 | TNFSF10:36 MYPN41 STAT1:219 OAS2:281 PLEKHA4:290 HCAR2:299                   |
| BLANCO_MELO_HUMAN_PARAINFLUENZA_VIRUS_3   | 0.003964052  | 81        | 9.020e-01 | 9.986e-01 | PRODHD:40 CCNS:79 SERPINAA6:139 EFCAB2L:225 CAMSPA3:433 CA9:467              |
| BLANCO_MELO_RESPIRATORY_SYNCYTIAL_VIRUS_3 | 0.004815896  | 194       | 9.598e-01 | 9.986e-01 | TNFSF10:36 STAT1:219 PLEKHA4:290 HCAR2:299 TNFRSF9:312 TNF:363               |
| BLANCO_MELO_RESPIRATORY_SYNCYTIAL_VIRUS_3 | -0.044030487 | 83        | 1.635e-01 | 9.986e-01 | MCFL2:317 PAD2:348 MB:354 DACT2:430 HLA-DMB:453 SYNGR4:534                   |
| BLANCO_MELO_COVID19_SARS_COV_2_INFECTION  | -0.038275814 | 59        | 3.096e-01 | 9.986e-01 | NABP3:176 DNMT1:7360 RASGRP2:389 SDR16C5:395 INHBE:622 BCL2A1:759            |
| BLANCO_MELO_COVID19_SARS_COV_2_INFECTION  | -0.086149760 | 51        | 3.344e-02 | 9.986e-01 | KLHDC7A:195 VIL1:236 SLC25A10:324 DACT2:430 HLA-DMB:453 SYNGR4:534           |
| BLANCO_MELO_COVID19_SARS_COV_2_INFECTION  | 0.007984507  | 240       | 6.714e-01 | 9.986e-01 | TNFSF10:36 HDAC9:43 STAT2:129 OAS2:281 NEDD4:300 TNFRSF9:312                 |

## DisGeNET Top pathways by non-permutation

| Geneset                                  | stat        | num.genes | pval      | p.adj     | gene.vals                                                                      |
|------------------------------------------|-------------|-----------|-----------|-----------|--------------------------------------------------------------------------------|
| Lymphoma, T-Cell, Cutaneous              | 0.07515422  | 177       | 5.967e-04 | 9.712e-01 | TNFSF10:36 HDAC9:43 IFNG:93 FASLG:99 FOXP3:105 STAT2:129                       |
| Hepatitis                                | 0.07615196  | 201       | 2.134e-04 | 9.712e-01 | AFP:6 AKR1D1:27 TNFSF10:36 SLC06A1:72 IFNG:93 FASLG:99                         |
| Hepatitis C                              | 0.04794725  | 498       | 3.012e-04 | 9.712e-01 | AFP:6 AFM:31 TNFSF10:36 HDAC9:43 SLC06A1:72 IFNG:93                            |
| Psoriasis                                | 0.04585194  | 522       | 4.089e-04 | 9.712e-01 | TNFSF10:36 SLC06A1:72 NCKAP1:92 IFNG:93 LRK2:96 FASLG:99                       |
| Scleroderma                              | 0.09146342  | 119       | 5.908e-04 | 9.712e-01 | ENG:3 IFNG:93 PRSS55:98 CENPC:103 COL1A1:148 DNMT1:193                         |
| Tumor Immunity                           | 0.10674431  | 88        | 5.533e-04 | 9.712e-01 | AFP:6 CD86:62 IFNG:93 FOXP3:105 IL33:167 APC:258                               |
| 46, XX Testicular Disorders of Sex Devel | 0.05307051  | 3         | 7.502e-01 | 9.739e-01 | NRS4A1:5662.5 RSP01:5662.5 NR0B1:5662.5 NA NA NA                               |
| 46, XY Disorders of Sex Development      | 0.05481569  | 12        | 5.109e-01 | 9.739e-01 | SRD5A2:201 NR5A1:5662.5 LHCGR:5662.5 PAICS:5662.5 MAP3K1:5662.5 HSD17B7:5662.5 |
| 46, XY female                            | 0.13710425  | 10        | 1.333e-01 | 9.739e-01 | SRD5A2:201 WNT4:516 NR0B1:5662.5 HSD17B3:5662.5 DMRT2:5662.5 DMRT1:5662.5      |
| 5,10-Methyltetrahydrofolate reductase    | 0.10658180  | 15        | 1.531e-01 | 9.739e-01 | HPGD5:330 AKAP9:382 ABCB1:461 SLC01B1:510 XRC1:905 SERPINE1:5662.5             |
| Abnormalities, Drug-Induced              | 0.05306987  | 2         | 7.949e-01 | 9.739e-01 | CAT:5662.5 EPHX1:5662.5 NA NA NA NA                                            |
| Abortion, Habitual                       | 0.05308034  | 4         | 7.131e-01 | 9.739e-01 | MTHFR:5662.5 F5:5662.5 VEGFA:5662.5 NA NA                                      |
| Achondrogenesis, type IB (disorder)      | 0.05308557  | 5         | 6.810e-01 | 9.739e-01 | SLC26A3:5662.5 SLC26A2:5662.5 SLC26A5:5662.5 SLC26A4:5662.5 DCN:5662.5 NA      |
| Acidosis, Lactic                         | 0.03216919  | 125       | 2.157e-01 | 9.739e-01 | ALDH7A1:119 TRMT10C:280 GPCB:344 TNF:363 TFAM:413 NDUFA11:455                  |
| ACROMESOMELIC DYSPLASIA, MAROTEAUX TYPE  | 0.05306987  | 2         | 7.949e-01 | 9.739e-01 | IGF1:5662.5 GDF5:5662.5 NA NA NA NA                                            |
| Acute Myeloid Leukemia, M1               | 0.04804585  | 76        | 1.453e-01 | 9.739e-01 | TNFSF10:36 FLT3:159 CD9:306 MPO:391 S100A8:392 AQP9:569                        |
| Acute myeloid leukemia, minimal differen | 0.08298290  | 15        | 1.465e-01 | 9.739e-01 | FLT3:159 MPO:391 ABCB1:461 NCR1:715 FAS:5662.5 FANCB:5662.5                    |
| Adenocarcinoma of lung, stage IV         | -0.02552503 | 11        | 7.695e-01 | 9.739e-01 | ROS1:1143 FBLN2:101 CD74:2287 TP53:7003.5 RET:7003.5 EGFR:70                   |